ML03 Arithmetic, converting text and reading errors

Week 1, Python and a first prediction | Session 3 of 24 | Machine Learning

Today's question. How do I turn the text of a price into a number I can calculate with?

Objective. By the end: you can calculate with the five basic operators, turn text into a number, and find the line an error points to.

> **WHERE THIS IS USED**
> Anything you type into a form arrives as text. The app has to turn it into a number before it can add, compare or charge. Where in the apps you use does text quietly become a number?

Before this session: ML02 Values, names and types. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
operator	A symbol that tells Python which calculation to do.	The plus and minus keys on a calculator.
parentheses	Brackets that make Python calculate the inside first.	Doing the part in brackets first in maths.
comment	A line starting with # that Python ignores and people read.	A note written in the margin.
convert	Change a value from one type to another, such as text to a number.	Changing shillings into dollars before adding.
error message	Python telling you what it could not do and where.	A friend pointing to the exact line that went wrong.
traceback	The list of lines Python followed before the error.	A trail of footprints that ends at the problem.

Block A. The basic operators

Run the cell. Notice the difference the brackets make.

Predict. Predict the four numbers before you run. Write your guess on paper first.

Your guess, written before you run it

```
price = 3000
total = price * 4
share = price / 4
no_brackets = price - 500 * 2
with_brackets = (price - 500) * 2
```

```
print(total, share, no_brackets, with_brackets)
```

Run it, then compare. The cell prints 12000, 750.0, 2000 and 5000. + adds, - subtracts, * multiplies and / divides. Multiplication happens before subtraction, so the brackets change the answer from 2000 to 5000. Brackets are called parentheses.

Investigate.

1. Why do 2000 and 5000 differ?
2. Why is share 750.0 and not 750?

Modify. Write the calculation for 3 books at UGX 3000 plus 2 pens at UGX 1000 in one line.

Block B. Text becomes a number

A price arrives as text from a form. Run the cell and read the result.

Predict. Predict the output. Can Python multiply text by 4 and give 50.0? Write your guess on paper first.

Your guess, written before you run it

```
# The price came in as text
text_price = "12.5"
price = float(text_price)
total = price * 4
print(total)
```

Run it, then compare. The cell prints 50.0. float turns the text 12.5 into a number. The line starting with # is a comment, which Python ignores. If the text were abc, float would stop with a ValueError and the last line of the error would name the problem.

Investigate.

1. Which line of an error message do you read first?
2. What does float("12,5") do?

Modify. Change the text price to "20" and run the cell again.

Make

Convert "12.5" to a number and calculate the price of four items. Predict the answer before you run it. Explain why nobody can divide by zero.

Save your work as ML03. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
ValueError: could not convert string to float	The text has letters, a comma or a currency symbol inside it.	Remove everything except digits and one decimal point.
ZeroDivisionError: division by zero	The calculation divides by zero.	Check the number you divide by; it must not be 0.

Exit ticket

float turns text such as 12.5 into a number, and an error message names the problem on its last line.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.

After this session: foundation lab F01 on the fourth day of the week.